Table 1: L2 (TDOST classes), all classes — comparable to TDOST / DeepCASAS; $m = 1$ labelled windows per cluster. W/M = weighted/macro F1.

Method	Repr.	K	Milan		Aruba		Cairo		Average	
			W-F1	M-F1	W-F1	M-F1	W-F1	M-F1	W-F1	M-F1
k-means	Sensor hist.	20	0.71	0.37	0.81	0.45	0.70	0.35	0.74	0.39
k-means	Sensor hist.	100	0.74	0.42	0.88	0.47	0.72	0.35	0.78	0.42
k-means	Sensor+ToD hist.	20	0.69	0.28	0.76	0.29	0.81	0.36	0.76	0.31
k-means	Sensor+ToD hist.	100	0.71	0.40	0.86	0.46	0.84	**0.46**	0.80	0.44
k-means	MLM	20	0.74	0.32	**0.89**	**0.51**	0.72	0.27	0.78	0.37
k-means	MLM	100	0.77	**0.46**	0.85	0.48	0.77	0.40	0.80	0.45
k-means	MLM-time	20	0.74	0.32	0.84	0.41	0.74	0.26	0.78	0.33
k-means	MLM-time	100	0.74	0.41	0.85	0.47	0.74	0.31	0.78	0.40
k-means	MLM + ToD	20	0.63	0.26	0.80	0.32	**0.86**	0.44	0.76	0.34
k-means	MLM + ToD	100	**0.79**	0.41	0.82	0.44	**0.86**	**0.46**	**0.82**	0.44
k-means	MLM-time + ToD	20	0.73	0.29	0.71	0.29	0.82	0.42	0.75	0.33
k-means	MLM-time + ToD	100	0.76	0.44	0.84	0.46	**0.86**	**0.46**	**0.82**	**0.46**
SCAN	MLM	20	0.76	0.33	0.77	0.32	0.78	0.35	0.77	0.34
SCAN	MLM	100	0.74	0.40	0.85	0.43	0.73	0.35	0.78	0.39
SCAN	MLM-time	20	0.75	0.39	0.81	0.46	0.78	0.36	0.78	0.40
SCAN	MLM-time	100	0.76	0.39	0.83	0.43	0.78	0.41	0.79	0.41
DEC	MLM	20	0.76	0.36	0.88	0.46	0.73	0.31	0.79	0.38
DEC	MLM	100	0.75	0.37	0.87	0.45	0.74	0.32	0.79	0.38
DEC	MLM-time	20	0.76	0.36	0.86	0.46	0.78	0.34	0.80	0.39
DEC	MLM-time	100	0.76	0.37	0.87	0.46	0.76	0.34	0.80	0.39

Table 2: L2, rare classes masked from the average; $m = 1$ labelled windows per cluster. W/M = weighted/macro F1.

Method	Repr.	K	Milan		Aruba		Cairo		Average	
			W-F1	M-F1	W-F1	M-F1	W-F1	M-F1	W-F1	M-F1
k-means	Sensor hist.	20	0.72	0.47	0.81	0.72	0.70	0.34	0.74	0.51
k-means	Sensor hist.	100	0.75	0.52	0.88	0.76	0.72	0.36	0.78	0.54
k-means	Sensor+ToD hist.	20	0.70	0.35	0.77	0.46	0.81	0.42	0.76	0.41
k-means	Sensor+ToD hist.	100	0.71	0.50	0.86	0.74	0.84	0.47	0.81	0.57
k-means	MLM	20	0.75	0.40	**0.89**	**0.81**	0.72	0.32	0.79	0.51
k-means	MLM	100	0.77	**0.58**	0.86	0.76	0.78	0.42	0.80	0.58
k-means	MLM-time	20	0.75	0.40	0.84	0.65	0.75	0.31	0.78	0.45
k-means	MLM-time	100	0.74	0.51	0.85	0.75	0.75	0.36	0.78	0.54
k-means	MLM + ToD	20	0.63	0.32	0.80	0.51	**0.86**	0.52	0.77	0.45
k-means	MLM + ToD	100	**0.79**	0.51	0.82	0.71	**0.87**	**0.54**	**0.83**	0.58
k-means	MLM-time + ToD	20	0.73	0.36	0.72	0.46	0.83	0.49	0.76	0.44
k-means	MLM-time + ToD	100	0.77	0.54	0.84	0.74	**0.87**	**0.54**	**0.82**	**0.61**
SCAN	MLM	20	0.76	0.41	0.77	0.51	0.78	0.41	0.77	0.45
SCAN	MLM	100	0.74	0.51	0.86	0.68	0.74	0.41	0.78	0.53
SCAN	MLM-time	20	0.76	0.49	0.82	0.72	0.78	0.41	0.78	0.54
SCAN	MLM-time	100	0.77	0.48	0.83	0.69	0.78	0.44	0.79	0.54
DEC	MLM	20	0.76	0.45	0.88	0.74	0.73	0.36	0.79	0.52
DEC	MLM	100	0.75	0.46	0.87	0.73	0.75	0.37	0.79	0.52
DEC	MLM-time	20	0.77	0.45	0.86	0.73	0.78	0.40	0.81	0.53
DEC	MLM-time	100	0.77	0.46	0.88	0.74	0.76	0.39	0.80	0.53

Table 3: L1 (raw CASAS activities), rare classes masked from the average; $m = 1$ labelled windows per cluster. W/M = weighted/macro F1.

Method	Repr.	K	Milan		Aruba		Cairo		Average	
			W-F1	M-F1	W-F1	M-F1	W-F1	M-F1	W-F1	M-F1
k-means	Sensor hist.	20	0.70	0.46	0.82	0.63	0.18	0.20	0.57	0.43
k-means	Sensor hist.	100	0.74	0.55	0.87	0.63	0.27	0.24	0.63	0.48
k-means	Sensor+ToD hist.	20	0.69	0.36	0.76	0.35	0.67	0.35	0.71	0.35
k-means	Sensor+ToD hist.	100	0.70	0.50	0.86	0.61	**0.75**	**0.50**	0.77	**0.54**
k-means	MLM	20	0.73	0.38	**0.89**	**0.66**	0.21	0.20	0.61	0.41
k-means	MLM	100	0.76	**0.59**	0.85	0.64	0.35	0.31	0.66	0.51
k-means	MLM-time	20	0.73	0.41	0.84	0.55	0.29	0.16	0.62	0.37
k-means	MLM-time	100	0.73	0.53	0.86	0.63	0.31	0.28	0.63	0.48
k-means	MLM + ToD	20	0.62	0.29	0.79	0.35	0.73	0.46	0.72	0.37
k-means	MLM + ToD	100	**0.78**	0.53	0.82	0.59	0.74	0.49	**0.78**	**0.54**
k-means	MLM-time + ToD	20	0.72	0.34	0.71	0.38	0.71	0.44	0.71	0.38
k-means	MLM-time + ToD	100	0.75	0.54	0.84	0.62	0.73	0.44	0.77	**0.53**
SCAN	MLM	20	0.76	0.43	0.77	0.47	0.31	0.26	0.62	0.38
SCAN	MLM	100	0.74	0.51	0.85	0.56	0.30	0.28	0.63	0.45
SCAN	MLM-time	20	0.75	0.52	0.82	0.60	0.27	0.23	0.61	0.45
SCAN	MLM-time	100	0.76	0.50	0.83	0.53	0.32	0.32	0.64	0.45
DEC	MLM	20	0.76	0.47	0.88	0.61	0.20	0.22	0.61	0.43
DEC	MLM	100	0.74	0.47	0.87	0.60	0.30	0.26	0.64	0.44
DEC	MLM-time	20	0.76	0.49	0.86	0.62	0.31	0.26	0.65	0.46
DEC	MLM-time	100	0.75	0.49	0.87	0.62	0.29	0.27	0.64	0.46